COMMERCIAL ITEM DESCRIPTION

GENERATOR, SIGNAL
10 KHz to 1 GHz

The General Services Administration has authorized the use of this Commercial Item Description.

Scope. This commercial item description (CID), describes a laboratory standard quality bench piece of electronic test equipment which is a bench mount with rackmount option, solid-state, general purpose, 10 KHz to 1 GHz, generator signal. This CID is meant as a minimum requirement for the generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be type III, class 5, style E and color R in accordance with MIL-T-28800. This CID makes provision for the bid sample testing in accordance with the latest MIL-T-28800.

Applicable documents.

Specifications and standards. The following specifications and standards, form a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for (the latest edition).

STANDARDS

MILITARY

MIL-STD-461	Electromagnetic/Emission and Susceptibility Requirements for the Control of Electromagnetic Interface.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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Other publications. The following document forms a part of this CID to the extent specified herein. Unless otherwise indicated, the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS

IEEE Std 587-1980

(Application for copies of IEEE publications should be addressed to the Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017)

Order of precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Salient characteristics. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

1. Safety. The equipment shall be UL (Underwriters Laboratories) listed and approved or shall comply with the type III safety requirements of the latest MIL-T-28800.

a. Leakage current protection. Protection shall be provided from leakage current in excess of 3.5 milliamperes (ma) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es).

b Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated for at least four (4) times the operating voltage.

c. Type III ventilation openings.

(1) Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts.

(2) Top ventilation openings shall incorporate small mesh screening or similar design to prevent the expulsion of hot electronic particles due to failure of electronic devices such as capacitors.

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2. Parts, materials, and processes. Except for marking, the generator signal shall comply with the type III parts, materials, and process requirements of MIL-T-28800. Nonstandard parts approval requests shall be in accordance with MIL-T-28800. Front panel marking shall be in accordance with MIL-T-28800.

a. Restricted material. The equipment shall comply with the restricted material requirements of the latest MIL-T-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3. Design and construction. Unless otherwise specified herein, the equipment shall comply with the Type III design and construction requirements of the latest MIL-T-28800. The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits where construction is susceptible to leakage or corona, indicators and associated controls shall be in close proximity to each other, all potentiometer controls shall have at least a 270 degree rotation range (4.71 radians).

a. Type of construction. Construction of the generator shall be of a monolithic design.

b. Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be type BNC female, except for RF output connector(s) which may be type N.

c. Internal cooling. The equipment shall be capable of operating continuously for twenty four(24) hours at 50°C. If forced air is used, it shall be as specified in MIL-T-28800.

(1) Forced air cooling motor. DC brushless motors shall be utilized for the forced air cooling motor in the equipment.

(2) Air filter. Equipment utilizing a cooling scheme which exhausts air from the equipment through the cooling fan shall not require an externally accessible air filter.

(3) Thermal switch. A thermal switch shall be provided which will interrupt circuit power to the generator when the internal temperature rises to the degree of possible damage to the electronic components. A minimum of 10°C hysteresis drop from this tripping temperature must occur before circuit power is again applied.

4. Electrical power sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 120, 220 and 240 volts ac (±10%) at line frequencies of 50 Hz to 60 Hz.

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a. Electrical power connectors. The equipment shall operate with a ground the primary source on either side of the single phase input.

b. Power consumption. Maximum power consumption shall not exceed 450 volt ampheres V.A.

c. Fuses. Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements.

d. Surge and transient voltage protection. The generator shall contain surge and transient voltage protection internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per type III solid state tolerance bands of MIL-T-28800, the surge voltage protection circuit shall conform to IEEE Std 587-1980. The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection. Normal operation of the generator may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 percent and frequency transient conditions greater than ±5 percent from within the nominal type III steady-state tolerance bands of MIL-T-28800, damage (other than fuse failure) to the generator shall not occur during voltage transient conditions greater than ±20 percent and frequency transient conditions greater than ±5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) specified in MIL-T-28800. Normal operation of the generator shall be automatically resumed when the transient has passed and any blown fuses are replaced. Blown fuses shall be externally accessible. Testing shall be in accordance with MIL-T-28800.

5. Dimensions and weight. The size and weight of the equipment shall be consistent with current commercial capabilities, refer to paragraphs 5a and 5b of this CID.

a. Equipment dimensions. The nominal dimensions for the generator should be approximately 9 inches height x 17 inches wide x 24 inches deep.

b Equipment weight. The weight including all accessories (excluding manual), shall not exceed 90 lbs.

6 Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the style E enclosure requirements of MIL-T-28800.

a. Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

b. Tilt-bail handle. A tilt-bail handle may be provided in lieu of the hinged metal handles in accordance with MIL-T-28800.

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c. Reusable pouch or container. An attached accessory pouch may be used for storage of all supplied accessories in lieu of front-cover storage.

7. Marking and identification.

a. Marking. The visible surface adjacent to panel facilities such as variable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation thereof, indicating the purpose or use of the part.

b. Panel marking. The front panel marking shall comply with the type III parts, materials and processes panel marking requirements of MIL-T-28800; however, marking identification processes other than the type III requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with MIL-T-28800.

c. Items to be marked.

(1) Volume and weight (including accessories) shall be marked on three external surfaces of the enclosure.

(2) The type designation of the AN nomenclature of the generator shall be stenciled on the exterior or main frame on at least three adjacent surfaces, on the transient case on two opposite sides. Marking shall be in block letters approximately 2 inches in height.

8. Environmental requirements.

a. Temperature requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class 5 temperature requirements specified by MIL-T-28800 and as listed below.

(1) Non-operating temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -15°C to +60°C without loss of memory, -40°C to +70°C normal.

(2) Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0°C to 50°C.

b. Humidity requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for class 5 equipment as specified by MIL-T-28800.

c. Altitude requirements. The equipment shall meet its required performance and accuracy requirements under the altitude requirements for class 5 equipment as specified by MIL-T-28800.

d. Vibration requirements. The equipment shall meet the class 5 vibration Requirements as specified by MIL-T-28800.

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e. Mechanical shock requirements.

(1) Shock, pulse. The equipment shall meet the class 5 shock, pulse requirement as specified by MIL-T-28800.

(2) Transit drop. The equipment shall meet transit drop requirement as specified by MIL-T-28800.

(3) Bench handling. The equipment shall meet the requirements as specified by MIL-T-28800.

(4) Shock, high impact. The equipment shall meet the requirements as specified by MIL-T-28800.

f. Electromagnetic interference (EMI). The equipment shall comply with the following EMI requirements specified in MIL-STD-461:

(1) CE01: Narrow band requirement from 10kHz to 20 kHz
(2) CS03. Full limits.
(3) CS01: Full limits.
(4) CS02· Full limits.
(5) CS06 Full limits.
(6) RE01: Relaxed 10dB at the fundamental, 3rd harmonic, and 5th harmonic of the source power frequency with the measurement probe located 12 inches from device under test.
(7) RE02· Full limits to 10 times the highest used or intentionally generated frequency, or GHz, whichever is greater.
(8) RS01 Full limits.
(9) RS03: Full limits to 1GHz.

g. Magnetic environment. The equipment shall comply with the magnetic environment requirements of MIL-T-28800.

9. Reliability. Reliability verification testing, shall be in accordance with the type III, class 5 environmental conditions of MIL-T-28800.

a. Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be 2,000 hours.

b. Pre-conditioning requirement. Each deliverable equipment shall be subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being also accomplished in the sample testing phase.

c. Burn-in. Burn-in shall expose the equipment to extended periods of elevated temperatures, (50°C for a minimum of 96 hours). Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of MTBF. Prior to and following burn-in, a satisfactory operation test (SOT) shall be performed. A satisfactory operation check (SOC) shall be conducted at selected times during the burn-in test.

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d. Thermal cycling. Thermal cycling shall expose the equipment to hot and cold cycles for class 5 equipment per MIL-T-28800. Each equipment shich fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the dydling, a satisfactory operation check (SOC) shall be conducted during the last two upper and lower temperature limits.

10. Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules.

a. Maintainability. The equipment shall meet the type III maintainability requirements specified by MIL-T-28800.

11. Interchangeability. The equipment shall meet the type III interchangeability requirements specified by MIL-T-28800.

12. Performance characteristics. The generator shall be MATE compatible with both CIIL and ASCII controllable language through an external IEEE 488 bus configuration. The CIIL translation (TMA) shall be integral to the box where available or shall be supplied with the unit as a one unit purchase item when supplied in a separate box configuration.

a. Frequency. The minimum frequency range of the generator shall be 10 KHz to 1 GHz.

(1) Resolution. The frequency resolution of the generator shall be a maximum of 1 Hz throughout the entire frequency range.

(2) Accuracy. The frequency accuracy shall be based on the internal reference oscillator aging rate which shall be 4×10^{-9} per day after 3 months of continuous operation.

b. Spectral purity.

(1) Harmonics. The generator harmonics shall be -25 dBc or greater over the entire frequency range.

(2) Sub-harmonics. There shall be no noticeable sub-harmonics to 1 GHz.

(3) Spurious non-harmonics. The spurious non-harmonics shall be at least -80 dBc over entire frequency spectrum.

(4) Power line related or microphonically generated. The 60 Hz power line or 300 Hz microphonically generated effect shall be at least -80 dBc to 320 MHz and at least -70 dBc from 320 MHz to 1 GHz.

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(5) Absolute SSB phase noise. The absolute single sideband (SSB) phase noise at 200 MHz carrier at 20 KHz offset shall be at least -130 dBc per Hz.

c. Output. The minimum output level range into 50 ohm nominal shall be +16 dBm to -140 dBm.

(1) Output resolution. The generator output resolution shall be 0.1 dB.

(2) Standing wave ratio. The standing wave ratio shall be less than 2.0:1 for output levels of 0 dBm or greater.

(3) Output flatness. The output flatness shall be equal to or less than ±0.75 dB over entire range at the +10 dBm level.

(4) Output protection. The generator shall have reverse power protection of a minimum of 25W at +50 VDC from a 50 ohm source.

d. Amplitude modulation. The amplitude modulation depth shall be 0 to 99% up to an output level of +10 dBm.

(1) AM resolution. The AM resolution shall be 1%.

(2) Accuracy. The AM accuracy at the 1 KHz rate shall be better than ±(5% of setting +1% AM).

(3) Distortion. The generator AM distortion at the 1 KHz rate shall be less than 2% up to 30% AM and less than 6% up to 80% AM.

(4) Incidental phase modulation. The incidental phase modulation at the 1 KHz rate and 30% AM shall be less than 0.2 radians peak.

e. Frequency modulation and coupling. The modulation bandwidth at 3 dB down shall be a minimum of 20 Hz to 100 KHz AC coupled. DC to 100 KHz DC coupled internal or external into 600 ohms.

(1) Incidental AM with FM. The incidental AM with FM shall be less than 0.3% for a deviation of 20 KHz at a 1 KHz rate.

(2) FM distortion. The generator shall have a distortion of less than 4% for maximum dc coupled deviation and less than 2% for 50% maximum deviation.

(3) Accuracy. The FM indicator accuracy shall not exceed ±(5% of setting +10 hz) maximum.

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(4) Resolution. The FM resolution shall be less than 1.0% of setting or 0.0005% of maximum deviation whichever is larger.

(5) Deviation. The maximum deviation at 1 GHz shall be DC - 100 KHz.

f. Phase modulation. The maximum phase deviation at 1 GHz shall not be greater than 200 radians.

(1) Accuracy. The deviation accuracy shall not be greater than 10% at the 1 KHz rate.

(2) Bandwidth. The minimum 3 dB phase modulation bandwidth shall be 20 Hz to 10 KHz.

(3) Resolution. The synthesized signal generator phase modulation resolution shall be within 0.8% of setting or 0.005% of maximum deviation, whichever is greater.

g. Pulse modulation. The on/off ratio for the pulse modulation shall be 40 dB or greater.

(1) Rise and fall time. The rise and fall time between 10% and 90% of the pulse shall be 0.4 microseconds or less.

(2) Repetition rate. The minimum repetition rate for a carrier frequency of 10 MHz to 1 GHz shall be 20 Hz to 100 KHz AC and DC to 100 KHz DC.

(3) Pulse width. The minimum pulse width shall be at least 2 microseconds.

(4) Input level. The nominal peak input threshold shall be 1.5V.

h. Internal modulation. The generator shall have an internal modulation source of a minimum of 400 Hz and 1 KHz with a minimum output level of 2V rms ±0.5 dB into 600 ohms.

i. Digital sweep. The generator shall have a variable 10 KHz to 1 GHz front-panel selectable sweep with selectable sweep speed.

j. Manuals. The unit shall come with two(2) complete sets of operation, instruction and programming manuals.

13. Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

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Preservation, packaging, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. (The Government reserves the right to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract).

Custodian.
Air Force - 99
Navy - EC
DLA -ES

Review activity:
Army - CR

Preparing activity:
Air Force - 82

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